

Walking the hierarchy

last-point-of-contact Markov kernels, their exact spectra,
and what a walk can do for activation means

Working note VII for the continuation-algebra programme
self-reviewed draft, not independently verified

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Abstract

The expander property, stated plainly: the average of node values along a short random walk approximates their mean, with explicit bounds. We build its analogue on a Cantor set, as proposed.

- The groupoid is the pair groupoid of the Cantor set, filtered by “same cylinder” relations.
- The functions on arrows are kernels $\psi(x \wedge y)$ that depend only on the *last point of contact* $x \wedge y$ of two paths in the Michon tree, the same vertex that defines the ultrametric $d(x, y) = \kappa(x \wedge y)$.
- A nonnegative normalized kernel is one Markov step, convolution powers are walks, and the walks are Pearson–Bellissard-type processes on the Cantor set.

(1) Exact spectrum (any tree, any masses). Every such kernel is diagonal in the hierarchical Haar decomposition. On the detail space of vertex v its eigenvalue is

$$\lambda_v = 1 - \Pr(\text{contact strictly above } v) - \psi(v) \mu[v].$$

Conversely, a spectrum comes from a Markov kernel iff an explicit recursion for ψ stays nonnegative, so the best walk for given data is a convex program. The formula is verified to 10^{-11} on a random tree.

(2) Explicit walk bounds. The exact variance of walk averages, and Hoeffding tails with factor $(1 - \lambda^+)/(1 + \lambda^+)$.

(3) What a walk cannot do. A walk with nonnegative spectrum (every expander walk, every heat kernel) is never more accurate than independent sampling. It saves random bits, not forward passes. Gains need *negative* eigenvalues, i.e. kernels that jump into a sibling subtree, and are capped by the largest single-level share of variance. The tail groupoid’s stratification removes all coarse levels exactly.

(4) The weights enter through κ . If the activation is L -Lipschitz for κ and the Cantor set has box dimension D in κ , the error is $O(LT^{-1/2-1/D})$, and this rate is optimal. Choosing κ from the weights means choosing a hierarchy in which D is small.

(5) The verdict for random ReLU networks. Gaussian inputs admit exact fair Cantor codings along any orthonormal frame, so all of this is implementable. Measured at width 256, stratification gains are:

- random frames: $\approx 1.00\times$;
- a frame from the weights alone (the linearized network): $1.30\times, 1.06\times, 1.02\times$ at depths 4, 8, 16;
- pilot-estimated Stein/Hessian frames: $1.51\times, 1.33\times, 1.15\times$.

The mechanism is real but its payoff is small, because a deep random network’s variance does not live at coarse levels of any cheap hierarchy. In the mean-field limit it sits at Hermite degree $\sim L^2$, by note IV’s critical Galton–Watson law.

1 The construction, made exact

Tree, Cantor set, contact. Let T be a rooted, locally finite tree with no leaves (a Michon tree) and $\Omega = \partial T$ its space of infinite paths, a Cantor set. For $x \neq y$, the *last point of contact* $x \wedge y$ is the deepest common vertex. A decreasing weight $\kappa : V \rightarrow (0, \infty)$ with $\kappa \rightarrow 0$ along paths defines the ultrametric $d(x, y) = \kappa(x \wedge y)$; by Michon's theorem every ultrametric Cantor set arises this way (Bellissard–Julien). A probability μ with $\mu[v] > 0$ fixes cylinder masses. Write $\mathcal{F}_n$ for the σ -algebra of depth- n cylinders, E_n for conditional expectation, and $\text{sh}_u(x) = [u] \setminus [\text{child of } u \text{ on } x]$ for the shell of u seen from x .

Groupoid and convolution. The pair groupoid $\Omega \times \Omega$ is filtered by the closed subgroupoids $R^{(n)} = \{|x \wedge y| \geq n\}$, “same depth- n cylinder”. Its convolution algebra, with μ as Haar system, consists of kernels with $(k \star l)(x, y) = \int k(x, z) l(z, y) d\mu(z)$, acting on $L^2(\mu)$ by $Kf(x) = \int k(x, y) f(y) d\mu(y)$.

- A kernel is *radial* if $k(x, y) = \psi(x \wedge y)$ for a function $\psi \geq 0$ on vertices.
- It is *Markov* if $\int k(x, y) d\mu(y) = 1$, i.e. $\sum_{u \in \text{path}(x)} \psi(u) \mu(\text{sh}_u(x)) = 1$; on a finite-depth tree, holding at the leaf is the deepest contact.
- The t -fold convolution power is then the t -step transition kernel. The process jumps from x to y with probability determined by *where their paths separate*.

The conditional expectations E_n (kernel $\mathbf{1}\{|x \wedge y| \geq n\}/\mu[x]_n$) and the Pearson–Bellissard heat semigroups $e^{-t\Delta_s}$ are in this class.

The *tail groupoid*, whose arrows change a finite prefix and keep the tail, acts by bisections, i.e. inner symmetries (notes I and IV). It will supply the stratified schemes of §3.

2 The exact spectrum

For a vertex v with children $c_1, \dots, c_b$, the *detail space* D_v consists of the functions $\sum_i g_i \mathbf{1}_{[c_i]}$ with $\sum_i \mu[c_i] g_i = 0$. Then $L^2(\mu) = \mathbb{C} \oplus \bigoplus_v D_v$ orthogonally (martingale differences).

Theorem 2.1 (spectrum of last-point-of-contact kernels). *Let K be a radial Markov kernel.*

- (a) K is μ -reversible.
- (b) Each D_v is an eigenspace, with eigenvalue

$$\lambda_v = 1 - \bar{A}_v - \psi(v) \mu[v], \quad \bar{A}_v = \sum_{u \text{ strict ancestor of } v} \psi(u) \mu(\text{sh}_u(v)).$$

Here $\bar{A}_v$ is the probability that a step from a point of $[v]$ exits $[v]$, and it does not depend on the point.

- (c) Conversely, let λ be given on vertices. Define $\psi(\text{root}) = 1 - \lambda_{\text{root}}$ and, for a child v' of v ,

$$\psi(v') = \psi(v) + \frac{\lambda_v - \lambda_{v'}}{\mu[v']}.$$

Then the operator equal to λ_v on each D_v is a radial Markov kernel iff $\psi \geq 0$ (and leaf holding ≥ 0). In particular, spectra that decrease with depth always qualify when $\lambda_{\text{root}} \leq 1$.

(d) Radial kernels form a commutative algebra, and the optimal radial walk for given data (e_v) is a convex program: minimize $\sum_v e_v \frac{1+\lambda_v}{1-\lambda_v}$ subject to the linear constraints of (c).

Proof. (a) holds because $\psi(x \wedge y) = \psi(y \wedge x)$.

(b) Take $x \in [c_i]$ and $g \in D_v$. Split the integral over the shells of the vertices u on the path of x :

- for u strictly below v , the shell lies in $[c_i]$, where $g \equiv g_i$;
- for $u = v$, $\int_{[v] \setminus [c_i]} g d\mu = -\mu[c_i]g_i$;
- for u above v , the shell misses $[v]$ and contributes 0.

Using the Markov normalization, the contribution of the shells strictly below v is $1 - \bar{A}_v - \psi(v)(\mu[v] - \mu[c_i])$. Hence

$$Kg(x) = g_i \left[1 - \bar{A}_v - \psi(v)(\mu[v] - \mu[c_i]) - \psi(v)\mu[c_i] \right] = g_i [1 - \bar{A}_v - \psi(v)\mu[v]],$$

which is independent of i .

(c) Subtracting the formula of (b) for v' from that for v gives $\lambda_v - \lambda_{v'} = \mu[v'](\psi(v') - \psi(v))$, and $\lambda_{\text{root}} = 1 - \psi(\text{root})$. Positivity of the kernel is exactly $\psi \geq 0$.

(d) The operators are simultaneously diagonal. The objective is convex in λ , since $\lambda \mapsto (1 + \lambda)/(1 - \lambda)$ is convex on $[-1, 1)$, and the constraints are linear in λ . $\square$

For a fair binary tree with depth law $a_n = \Pr(\text{contact at depth } n)$, this reads $\lambda_m = A_{>m} - a_m$. For a fair b -ary tree it reads $\lambda_m = A_{>m} - a_m/(b - 1)$. A kernel that always jumps exactly at depth m (“into a sibling subtree”) has eigenvalue $-1/(b - 1)$ at depth m .

Check (`code/tree_spectrum.py`). On a random tree (depth 4, branching 2–4, Dirichlet masses, 79 leaves) with random ψ , $\max \|Kg - \lambda_v g\| = 8 \times 10^{-12}$ over all detail vectors, and detailed balance holds to 10^{-18} .

3 Walk bounds, and what a walk can do

Write $f = \mu(f) + \sum_v f_v$ with $f_v \in D_v$ and $e_v = \|f_v\|^2$, so $\sum_v e_v = \text{Var } f =: E$.

Theorem 3.1 (sampling along the walk). *Let $X_0 \sim \mu$ and let (X_t) be the chain of a radial Markov kernel. Then*

$$\text{Var} \left(\frac{1}{T} \sum_{t < T} f(X_t) \right) = \frac{1}{T} \sum_v e_v g_T(\lambda_v), \quad g_T(\lambda) = \frac{1 + \lambda}{1 - \lambda} - \frac{2\lambda(1 - \lambda^T)}{T(1 - \lambda)^2}.$$

If f takes values in $[0, 1]$ and $\lambda^+ = \max(\sup_v \lambda_v, 0) < 1$, then

$$\Pr(\bar{f}_T - \mu f \geq \varepsilon) \leq \exp\left(-2T\varepsilon^2 \frac{1 - \lambda^+}{1 + \lambda^+}\right)$$

(León–Perron’s optimal Hoeffding bound for reversible chains).

Proof. Use $\text{Cov}(f(X_s), f(X_t)) = \sum_v e_v \lambda_v^{|s-t|}$ (Theorem 2.1) and sum the geometric series. $\square$

Check. For $T = 50$ on the random tree, the formula gives 0.01114 and simulation gives 0.01100; independent sampling would give 0.00563.

Theorem 3.2 (what walks can and cannot do). (a) No free lunch. If every $\lambda_v \geq 0$, then $g_T(\lambda_v) \geq 1$, so the walk average is never more accurate than T independent samples. This covers every expander walk and every heat kernel. Expansion saves random bits, not evaluations.

(b) Radial walks. On a fair b -ary tree, with e_m the total energy at depth m , every radial walk has asymptotic variance ratio

$$\geq 1 - \frac{2}{b-1} \max_m \frac{e_m}{E},$$

because $\lambda_m \geq -a_m/(b-1)$, $\sum a_m \leq 1$ and $\frac{1+\lambda}{1-\lambda} \geq 1+2\lambda$. Gains require variance concentrated at a few levels.

(c) Stratification (tail groupoid). Take one independent μ -random point in each of the b^n depth- n cylinders, i.e. randomize the tails over the tail groupoid's prefix orbit. The variance is $e_{\geq n}/b^n$ with $e_{\geq n} = \mathbb{E} \text{Var}(f \mid \mathcal{F}_n)$. Every level above n is removed exactly.

Proof. (a) $g_T(\lambda) = T^{-1} \sum_{s,t < T} \lambda^{|s-t|} \geq 1$ for $\lambda \geq 0$. (b) Tangent-line bound at 0 and $\sum_m (e_m/E) a_m \leq \max_m e_m/E$. (c) Standard: the cells are independent, each contributes $\mu(c)^2 \text{Var}(f \mid c)$, and $\mu(c) = b^{-n}$. $\square$

Theorem 3.3 (the weights enter through κ). Suppose $|f(x) - f(y)| \leq L \kappa(x \wedge y)$, and suppose that for every δ the cut $\Gamma_\delta = \{v : \kappa(v) \leq \delta < \kappa(\text{parent})\}$ has at most $C\delta^{-D}$ cylinders (upper box dimension D). Allocate $T_v = \lceil \mu[v]T \rceil$ points to $v \in \Gamma_\delta$, with $\delta = (C/T)^{1/D}$. The total cost is at most $2T$, and

$$(\mathbb{E} |\hat{\mu}_T(f) - \mu(f)|^2)^{1/2} \leq \frac{1}{2} L C^{1/D} T^{-1/2-1/D}.$$

If (Ω, κ, μ) is Ahlfors D -regular, no randomized algorithm with T evaluations does better than $c L T^{-1/2-1/D}$ on this Lipschitz class.

Proof. On $[v]$, the oscillation of f is at most $L\kappa(v)$, so $\text{Var}(f \mid v) \leq L^2 \kappa(v)^2/4$. Hence

$$\text{Var} \hat{\mu}_T = \sum_v \frac{\mu[v]^2 \text{Var}(f \mid v)}{T_v} \leq \frac{L^2 \delta^2}{4T}.$$

For the lower bound, use Bakhvalov's argument with ultrametric bumps. In an ultrametric, $L\delta \mathbf{1}_{[v]}$ is L -Lipschitz whenever $\kappa(\text{parent } v) \geq \delta$. Put random signs on $2T$ disjoint such bumps; the unseen half carries error of order $L\delta \cdot T^{1/2} \cdot T^{-1}$. $\square$

So the improvement exponent over Monte Carlo is exactly $1/D$, the inverse dimension of the input Cantor set *measured in a metric adapted to the network*. “Defining κ cleverly from the weights” means choosing the hierarchy so that the activation is Lipschitz with small constant and small D : Bellissard–Julien's Michon-tree dimensions, now as a computational exponent.

4 Gaussian inputs: exact codings, Mehler walks, and where the variance lives

Proposition 4.1 (exact fair Cantor coding). Let $Q = (q_1, \dots, q_d)$ be any orthonormal frame and $b \geq 2$. The map sending $X \sim N(0, I_d)$ to the b -quantile digits of $\langle q_1, X \rangle, \langle q_2, X \rangle, \dots$ (with further digits by refining quantiles, interleaved in any order) is a measure isomorphism onto a fair b -ary Cantor set, the Pearson–Bellissard fair Cantor set. Every cylinder has known mass and can be

sampled exactly by inverse CDFs. All of §§2–3 is therefore implementable exactly on network inputs. The free choice is $(Q, \text{order}, \text{resolution})$, and that is where the weights enter.

Proposition 4.2 (Mehler walks cannot help even degrees). *For the walks $X' = \rho X + \sqrt{1 - \rho^2} Z$ with $\rho \sim \nu$ on $[-1, 1]$ (Ornstein–Uhlenbeck steps, antithetic flips and their mixtures), the eigenvalue on Hermite degree m is $\lambda_m = \mathbb{E}\rho^m$. Hence $\lambda_{2k} \geq 0$, and by Theorem 3.2(a) no such walk reduces the variance carried by even degrees.*

Where the variance lives (mean-field). By Mehler, $\mathbb{E}[F(X)F(X_\rho)] = \sum_m \|F_m\|^2 \rho^m$. In the infinite-width limit the left side, normalized, is the L -fold composition of note IV’s correlation map ϱ , a probability generating function. So the Hermite energy profile of an output is the generation- L law of the critical Galton–Watson process with offspring generating function ϱ . Its offspring tail is $\asymp m^{-5/2}$: the coefficients of $\varrho'' = (\pi\sqrt{1 - c^2})^{-1}$ decay like $m^{-1/2}$. That places it in the $3/2$ -stable domain, with survival $\asymp L^{-2}$ (note IV), and conditioned on survival Z_L is of order L^2 . The variance therefore sits at Hermite degree $\sim L^2$, deep in any hierarchy, and coarse cells capture a vanishing share as depth grows. At finite width the low-degree share is larger, as measured below.

Measurement (`code/energy.py`). Width $d = n = 256$, He weights, 2^{20} inputs, eight live outputs per network. The table gives the stratification gain $\text{Var } F / \mathbb{E} \text{Var}(F \mid \text{cell})$, bias-corrected, with b bins per direction along the first K frame directions.

depth	frame	$b=2, K=6$	$b=2, K=12$	$b=4, K=3$	$b=4, K=6$	$b=8, K=4$
4	random	1.005	1.009	1.004	1.007	1.005
4	weights only (linearized net)	1.170	1.185	1.252	1.265	1.303
4	pilot Stein + Hessian	1.244	1.245	1.409	1.413	1.508
8	random	1.003	1.006	1.002	1.005	1.003
8	weights only	1.046	1.064	1.041	1.064	1.051
8	pilot Stein + Hessian	1.188	1.189	1.285	1.288	1.333
16	random	1.002	1.003	1.001	1.002	1.002
16	weights only	1.010	1.017	1.006	1.014	1.009
16	pilot Stein + Hessian	1.088	1.088	1.128	1.129	1.147

The cells have 64 or 4096 points. The “weights only” frame takes the row of $W_L \frac{1}{2} W_{L-1} \cdots \frac{1}{2} W_1$ for the output, then the top singular vectors of that product: κ built from the weights, with no samples. The pilot frame uses 2×10^4 extra samples to estimate $\mathbb{E}[XF]$ and the leading eigenvectors of $\mathbb{E}[(XX^\top - I)F]$. It is an optimistic reference: in practice its cost would count against the budget. For comparison, the antithetic per-evaluation gains are 1.19, 1.09 and 1.08 at depths 4, 8 and 16.

5 Verdict and what would change it

- *The theory is what was hoped for.* Groupoid convolution by last-point-of-contact kernels gives Markov walks on the Cantor set, their spectra are explicit (Theorem 2.1), and their sampling errors have explicit bounds (Theorem 3.1). The weights enter through κ , and the payoff exponent is $1/D$ (Theorem 3.3).
- *But a walk only matches independent sampling* unless it anti-correlates (Theorem 3.2). For the challenge’s metric (forward passes), the value of the whole construction equals the variance share at the coarse levels of a computable hierarchy. For deep random He networks that share is small and falls with depth: weight-only gains of $1.30 \times \rightarrow 1.06 \times \rightarrow 1.02 \times$ over depths 4, 8, 16. This is consistent with note II’s design ceiling and with the Galton–Watson picture.

- *What would change it:* an activation with low effective dimension D in some weight-computable ultrametric. That would be the case for trained networks with low-rank structure, for strongly anisotropic weights, or for observables that depend on few directions. The theory then predicts $T^{-1/2-1/D}$.

A Scripts

`code/tree_spectrum.py`: Theorem 2.1 on a random tree, reversibility, and the walk variance (Theorem 3.1) against simulation. `code/energy.py`: stratification gains on exact Gaussian Cantor codings for random, weight-defined and pilot frames.